

Lecture Notes 11

Open Economy Macroeconomics and the International Monetary System

Intermediate Macroeconomics, EC2211

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Contents and literature

- The U.S. dollar in the international monetary system
- The balance of payments: international capital flows and the current account balance
- Dynamics of the international investment position
- Tariffs and exchange rates

Literature:

- Jones (2024), chapters 19-20
- Miran (2024), "A user's guide to restructuring the global trading system"
- Bénassy-Quéré (2025), "Is the international monetary system 'unfair'?"
- Kohlscheen et al. (2025), "Macroeconomic impact of tariffs and policy uncertainty"
- Klitgaard (2025), "Why does the U.S. always run a trade deficit?"

A walk through Stephen Miran's paper

A walk through Stephen Miran's paper

1. The U.S. dollar is the international reserve currency
2. The Triffin dilemma
3. The financial account and the international investment position
4. The exorbitant privilege
5. The flip side of the exorbitant privilege
6. Do tariffs help improve the trade balance?
7. A Mar-a-Lago Accord

In covering these seven headings, we will encounter a number of key concepts and ideas relating to international macroeconomics

1. The U.S. dollar is the international reserve currency

"Because America provides the reserve assets to the world, there is demand for U.S. dollars and U.S. Treasury securities that is not rooted in balancing trade or in optimizing risk-adjusted returns."

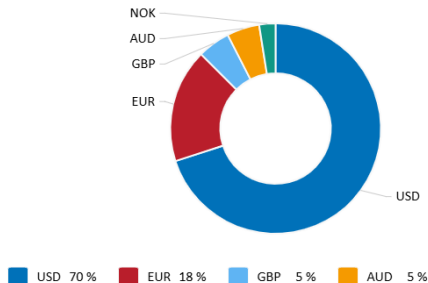
-Miran, p. 6

The foreign exchange markets

- Daily turnover USD 9.6 trillion
 - ... which is more than annual GDP in Germany and France combined
- U.S. dollar participated in 89% of trades, euro in 29%, yen 17%
- Largest trading centers: London, New York, Singapore

The U.S. dollar is the international reserve currency

- Used in international transactions
- Access to deep financial markets
- Reliable store of value
- ...



The Riksbank's FX reserves (SEK 425 billion in April 2025) mostly consists of U.S. dollars

What makes the U.S. dollar the international reserve currency?

- By far the largest financial market
- Large supply of liquid and safe assets (government bonds)
- Network effects
 - It is difficult to support several reserve currencies
 - ... just as it is difficult to support several "monies" in a country
 - The dollar is what you use for international transactions
 - ... and therefore the asset you hold in your liquidity reserves
- Rule of law, credible institutions, geopolitics, ...
 - Little risk of default and confiscation

Current question: Why don't the U.S. and Europe confiscate Russian reserve assets?

2. The Triffin dilemma

"In Triffin world, reserve assets are a form of global money supply, and demand for them is a function of global trade and savings, not the domestic trade balance or return characteristics of the reserve nation."

-Miran, p. 6

The Triffin dilemma

Triffin (1960s):

- The reserve currency country (the U.S.) must run *current account* deficits to supply the world with the reserve currency
- This means that the *net international investment position* of the United States deteriorates over time ...
- ... which in turn means that the dollar will not be able to remain a safe store of value

Miran:

- These forces have become stronger as:
 - The U.S. share of the world economy has diminished
 - The volume of international trade and transactions has increased more than global GDP

(Many economists however question the relevance of the Triffin dilemma in the modern economy – see e.g. [Bordo and McCauley](#))

The balance of payments

The **balance of payments** records all transactions between the home country and the rest of the world

It consists of three main headings:

- 1 The current account (commercial transactions)
- 2 The capital account (a minor thing that we can ignore)
- 3 The financial account (purchases and sales of assets)

The current account balance

The **current account balance**, in turn, is the sum of:

- 1 The trade balance
 - Net exports, NX
- 2 The income balance
 - Net factor income from abroad (interest and profits from abroad, labor income from abroad, ...)
- 3 Net unilateral transfers
 - In particular net foreign aid received

International capital flows: relating NX to CA to $S - I$

Recall the national income identity:

$$Y = C + I + G + NX$$

Let T denote taxation (net of transfers) and rewrite the identity as:

$$(Y - T - C) + (T - G) - NX = I$$

The first two terms on the left-hand side are:

- Private saving since $Y - T$ is disposable income and C is household spending
- Government saving since $T - G$ is the government's income minus spending

These two terms therefore add up to domestic saving, S

International capital flows: relating NX to CA to $S - I$

The national income identity can therefore be written as

$$NX = S - I$$

In other words, a country that runs a trade surplus ($NX > 0$), saves more than it invests domestically

This means that it saves abroad by accumulating foreign assets:

- There is a net capital outflow from this country
- ... meaning that domestic agents accumulate claims on foreign agents

International capital flows: relating NX to CA to $S - I$

What we just did follows Jones' presentation in chapter 19

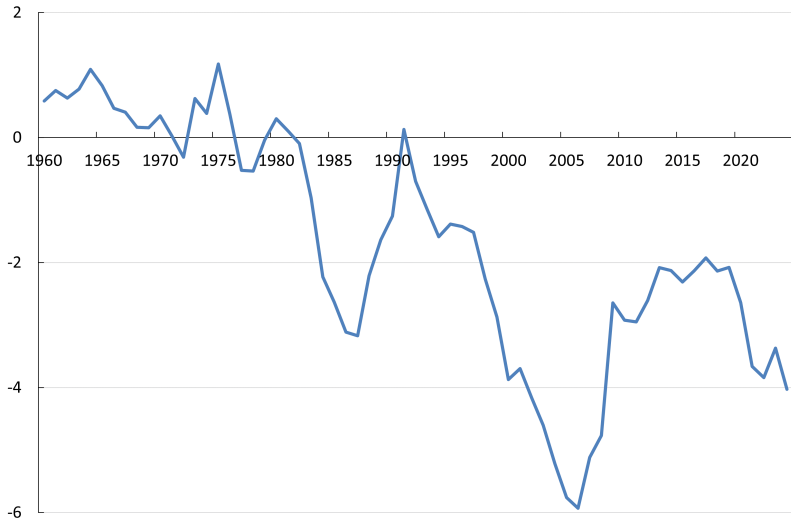
There is however an implicit assumption behind Jones' presentation:

- Jones assumes that there is no net factor income from abroad and no foreign aid
- In other words, he assumes that $CA = NX$ and that national income equals GDP

Allowing for net factor income, NFI , (and including foreign aid in NFI) the relations are:

$$CA = S - I = NX + NFI$$

The U.S. current account balance



Percent of GDP

The U.S. current account balance

The negative current account balance in the United States means that:

- The U.S. is saving less than what is invested in the country
- In other words: foreign capital flows in to the U.S. and contributes to its investments
- This also means that foreigners accumulate claims on the U.S.

Are persistent CA deficits sustainable?

Can a country, for example the United States, run a current account deficit forever?

To address this question, let us consider how the country's claims on the rest of the world develop over time

As we will see, the analysis is symmetric to the analysis in LN09 where we looked at government debt sustainability:

- The trade balance, NX , will play the same role as the primary budget surplus
- The current account balance, CA , will play the same role as the total budget surplus
- The **net international investment position**, $NIIP$, will play the same role as the government debt

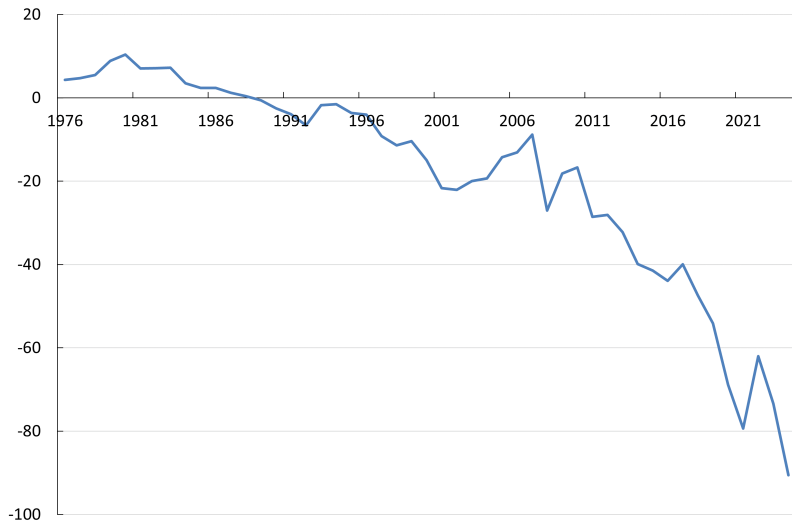
The international investment position and the current account balance

Let $NIIP$ denote the collective net claims of agents in the country on foreigners

If the country runs a current account surplus ($CA > 0$), it accumulates claims on foreigners:

$$NIIP_{t+1} = NIIP_t + CA_t \quad (1)$$

The U.S. net international investment position



Percent of GDP

The international investment position and the trade balance

Suppose that net factor income from abroad only consists of the return on the net investment position, and let i denote the interest rate:

$$NFI_t = i \times NIIP_t$$

Use this with $CA_t = NX_t + NFI_t$ to see that:

$$NIIP_{t+1} = NIIP_t + (NX_t + iNIIP_t) = (1 + i)NIIP_t + NX_t \quad (2)$$

Dynamics of the international investment position

We want to relate to GDP, so divide by Y_t in (1) and (2) to get:

$$(1 + g)niip_{t+1} = niip_t + ca_t$$

and

$$(1 + g)niip_{t+1} = (1 + i)niip_t + nx_t$$

where g denotes the GDP growth rate and $niip$, ca and nx denote these variables relative to GDP

Stabilizing the international investment position

From the previous slide, we see that $niip$ will remain constant if:

$$ca = g \times niip$$

and

$$nx = -(i - g) \times niip$$

Example: Suppose that $i = 5\%$, $g = 4\%$ and $ca = -4\%$

- The $niip$ will then stabilize at -100% of GDP, i.e. foreigners having claims on the U.S. amounting to 100 percent of annual GDP
- This would require $nx = -1\% \times niip$, i.e. that the U.S. runs a permanent trade *surplus* amounting to 1 percent of GDP
(We will return to this soon with some modifications)

3. The financial account and the international investment position

"In Triffin world, the reserve assets producer must run persistent current account deficits as the flip side of exporting reserve assets."

-Miran, p. 7

The financial account

Recall the balance of payments

The **financial account** is just the mirror image of the current account,
 $FA = CA$

Consider a current account surplus in Sweden

- Since $CA = S - I$, Sweden is saving more than it spends domestically
- If Swedish citizens save abroad by acquiring or building a new plant in China, this would be recorded as a "direct investments" in the financial accounts
- If Swedish citizens instead save abroad by purchasing U.S. government bonds, this would be recorded as a "portfolio investment" in the Swedish financial accounts ...
- ... but as a *negative* portfolio investment in the U.S. financial accounts

The financial account

So is the negative current account balance in the United States a consequence of the dollar's reserve status?

If foreigners build up reserves by purchasing U.S. assets, such as government bonds,

- there will be "outflows" on the U.S. financial accounts
- ... which must be matched by negative items on the U.S. current account balance

4. The exorbitant privilege

"Because there is a persistent reserve-driven demand for UST securities, the United States may be able to borrow at lower yields than would otherwise be the case."

-Miran, p. 8

The exorbitant privilege

The exorbitant privilege

Reserve currency status → high demand for U.S. treasury bonds → low borrowing costs for the U.S. government

- Term coined in 1960s by French government representatives
- The Bretton Woods system (1944 1971)
 - U.S. dollar pegged to gold
 - All other currencies pegged to USD
- France complained that U.S. fiscal policy was too expansionary
... and that France indirectly sponsored the Vietnam war
- The Bretton Woods system broke down but USD dominance has persisted

The exorbitant privilege

Recall the analysis of the dynamics of the NIIP

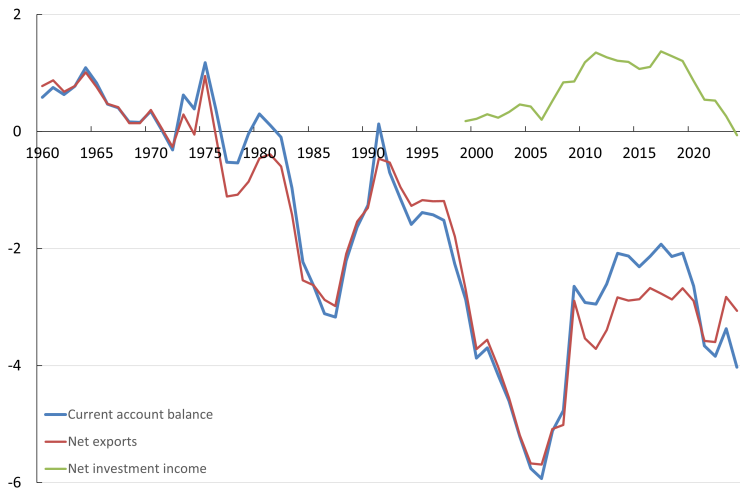
We concluded that the U.S. could continue running a CA deficit of 4 percent annually, but that it then must run a trade surplus to stabilize the NIIP relative to GDP

We assumed that $i = 5\%$

In reality, the U.S. foreign investments tend to generate a much higher return than the interest on its liabilities

Running persistent trade deficits may then be possible

The American CA, NX, and $i \times NIIP$



Percent of GDP. Note that in spite of foreigners having net claims on the U.S. amounting to almost 100 percent of GDP ($niip \approx -100\%$), the U.S. has had *positive* net returns on its foreign investment position!

5. The flip side of the exorbitant privilege

"America's status as reserve currency confers the burden of an overvalued currency eroding the competitiveness of our export sector"

-Miran, p. 11

6. Do tariffs help improve the trade balance?

"The critical question is to what extent currencies adjust to offset changes in international tax regimes."

-Miran, p. 13

How does the trade balance respond to tariffs?

Suppose that the United States impose large tariffs on its imports with the ambition to make imports more expensive and to stimulate its domestic manufacturing industry

Recall that

$$NX = EX - IM \approx S - I$$

Insight

The trade balance can only "improve" if either U.S. saving increases or investment falls

How does the trade balance respond to tariffs?

Not obvious that either S or I is affected by a tariff:

- In that case, the dollar needs to *appreciate* in response to the tariff
- The dollar appreciation partially offsets the higher import prices from tariffs
- ... and also reduces exports, leaving net exports unaffected
- The dollar did appreciate after Trump imposed tariffs in 2018
- ... but depreciated sharply in 2025 after new tariffs were announced

If $S - I$ is affected:

- Adjustment may come from lower investments in the United States
- ... which probably does not generate a revival for its manufacturing industry

The unwinding of the U.S. trade deficit

So tariffs are maybe mostly a source of tax revenue, rather than an instrument to rebalance U.S. international trade

Are there other tools to work with?

7. A Mar-a-Lago Accord

"Historically, multilateral currency accords have been the principal means of implementing international changes in the value of the dollar."

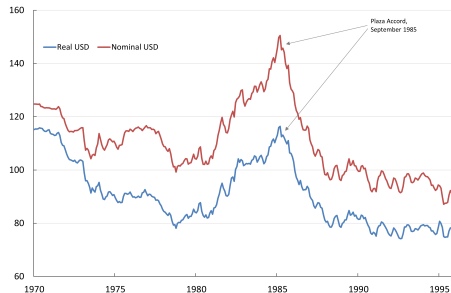
-Miran, p. 28

The Plaza Accord

- The dollar appreciated rapidly in early 1980s
- ... as both tight monetary policy and expansionary fiscal policy pushed up U.S. interest rates

The Plaza accord

- Meeting at Plaza Hotel in NY
- Large countries announced that "some further orderly appreciation of the non-dollar currencies is desirable"
- Again the "power of words" (and some minor interventions): the dollar depreciated substantially



USD exchange rate index (higher value = more appreciated dollar). Source: BIS.

The unwinding of the U.S. trade deficit

What if words are not enough?

"Economists tend to point to the Federal Reserve's policy rate as the main driver of the dollar."

"There is a variety of steps an Administration can take if it is willing to be creative, that do not rely on the Fed cutting rates."

-Miran, p. 30

The unwinding of the U.S. trade deficit

Miran's thoughts include:

- A "user fee" on foreign official holders of Treasury securities (i.e. withholding parts of interest payments)
- Secure "voluntary" cooperation from the Fed, including in capping interest-rate increases on government bonds
- Interventions on foreign exchange markets (buy foreign currency so that they appreciate)

See also exercise *9 in Jones chapter 20

Summing up

Key concepts covered:

- The balance of payments, the current account and the financial account
- The current account balance, the trade balance, saving and investment, international capital flows and the dynamics of the international investment position
- Tariffs and exchange-rate offset